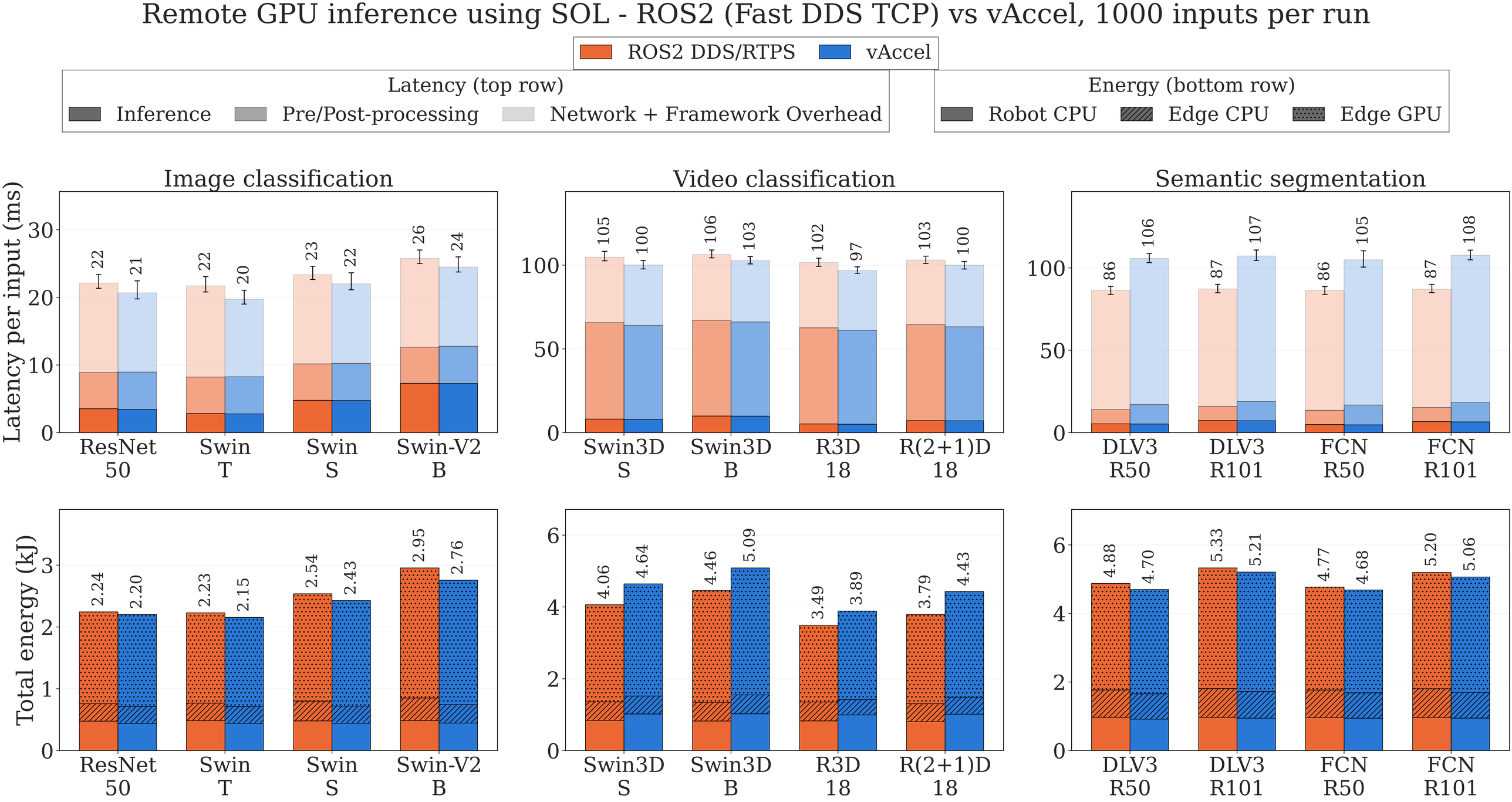

Remote GPU inference using SOL - ROS2 (Fast DDS TCP) vs vAccel, 1000 inputs per run
ROS2 DDS/RTPS
vAccel
Latency (top row)
Inference
Pre/Post-processing
Network + Framework Overhead
Energy (bottom row)
Robot CPU
Edge CPU
Edge GPU
Image classification
Video classification
Semantic segmentation
Latency per input (ms)
Total energy (kJ)
ResNet 50
Swin T
Swin S
Swin-V2 B
Swin3D S
Swin3D B
R3D 18
R(2+1)D 18
DLV3 R50
DLV3 R101
FCN R50
FCN R101
22
21
22
20
23
22
26
24
105
100
106
103
102
97
103
100
86
106
87
107
86
105
87
108
2.24
2.20
2.23
2.15
2.54
2.43
2.95
2.76
4.06
4.64
4.46
5.09
3.49
3.89
3.79
4.43
4.88
4.70
5.33
5.21
4.77
4.68
5.20
5.06